Easton Check In

May 19, 2026

1 Update

So basically, simple tests of species abundance and diversity wasn't really giving me clear results (not super surprising given the sample size) so I'm trying to fit GAMs to the data to try to parse out some sort of scientific story beyond 'here are the fish we found and the local names' as that's really all I have besides that.

I have some specific questions for you (below) but then what I could really use help on is just getting another pair of eyes on my models, my thought processes, and if there's anything useable here. As I've said before, I'm actually not really a stats person, just pretending to be so this is actually my first time trying to fit a GAM and write a stats paper. The purpose of this document is to walk you through my thinking and decisions so you can tell me if I made a wrong assumption somewhere. I was also following a lot of the advice that came up in this GAM tutorial: https://www.youtube.com/watch?v=sgw4cu8hrZM

Specific questions:

1. Looking for journal ideas. Can I use this UNH free open-access list you sent during the MADA meeting or no? https://quantmarineecolab.github.io/lab-manual/10-producing-quality-work.html#where-to-publish

2. In the models below, I include random effects on Desa (village) and site type (destroyed, replanted, and natural mangroves) because they were sampled repeatedly. Is that necessary as they were all sampled an equal amount (with the exception of one datapoint because we did lose one video in June)?

3. Some other data ideas for explanatory variables: I don't think there exists a daily temperature log for this region, and I'm mad I didn't bother to write this down. However, I may be able to pull it from the local airport, which is about an hour away. Another data idea is for one month, we measured % canopy cover for all of the sites, but we only did this one time, not repeatedly at each time we measured. Would that be worth trying to incorporate or does that not make sense statistically because it's just one point?

2 Intro to the data

Table 1: Response variables I'm testing

Variable	Description	Type	Distrbutions.Used
Total MaxN	Sum of all the MaxNs across species for each recording	Count	Poisson, Negative Binomial
Total Species	Total number of different species that showed up per recording	Count	Poisson, Negative Binomial
Shannon	Shannon's diversity index per recording	Continuous	Gaussian
Simpson	Simpson's diversity index	Continuous 0-1	Beta Regression

Table 2: Explanatory variables I'm considering

Variable	Description	Type	Spline.Used
Month	Note: I think time of dat and month are correlated because we went out at high tide every month, which changes the time of day we left. In each model I test which to use by comparing AIC scores	Ordered Categorical	Cyclic smooth
Time of day		Continuous	Nonlinear smooth
Village	Referred to as 'Desa'	Categorical	Fixed Effects factor
Habitat Type	D - Destroyed, E - EMR (replanted), L - Natural	Ordered Categorical	Ordered fixed effects factor
Visibility	Visibility of video. Not a variable but a correction	NA	
Vegitation	% of screen covered by vegitation	Continuous	Nonlinear smooth
Weather	Description of weather events: Sunny, Cloudy, Rainy	Ordered Categorical	Ordered fixed effects factor
Tide Height	Tide height at beginning of recording. From Fishing Points App. I think this is more accurate than the on-site depth measurements we took	Continuous	Nonlinear smooth

3 Testing for correlations

I think Month and Time of Day are correlated because we went out every month at high tide, which of course is at a slightly different time every month (see figure 2). Will expand on that at each model.

```
# Here I am combining some of the weather data and assigning numbers to months

idk <- cor(URUV_model_cont, method = "pearson", use="complete.obs")# cor(URUV_model_cont, method = c("p

corrplot(idk, type = 'upper', order = "hclust",
         tl.col = "black", tl.srt = 45)
```

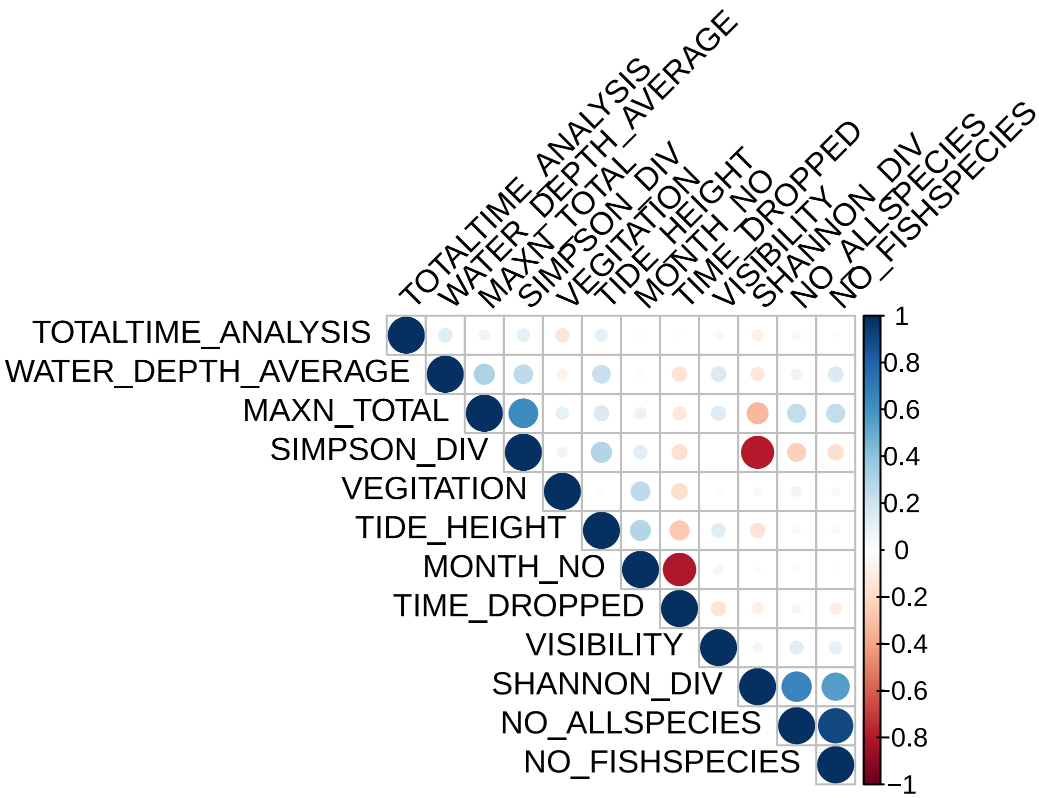


Figure 1: checking correlations between data variables

```
## The pearsons coeff for Tide height and time of day is -0.26546621 and time of day vs avg water depth

# WARNING I THINK MONTH AND TIME DROPPED ARE COLLINEAR WHICH MAKES SENSE BECAUSE WE PRETTY MUCH HAD A D
```

```
ggplot(URUV_model, aes(x = MONTH, y = TIME_DROPPED)) +
  geom_boxplot()
```

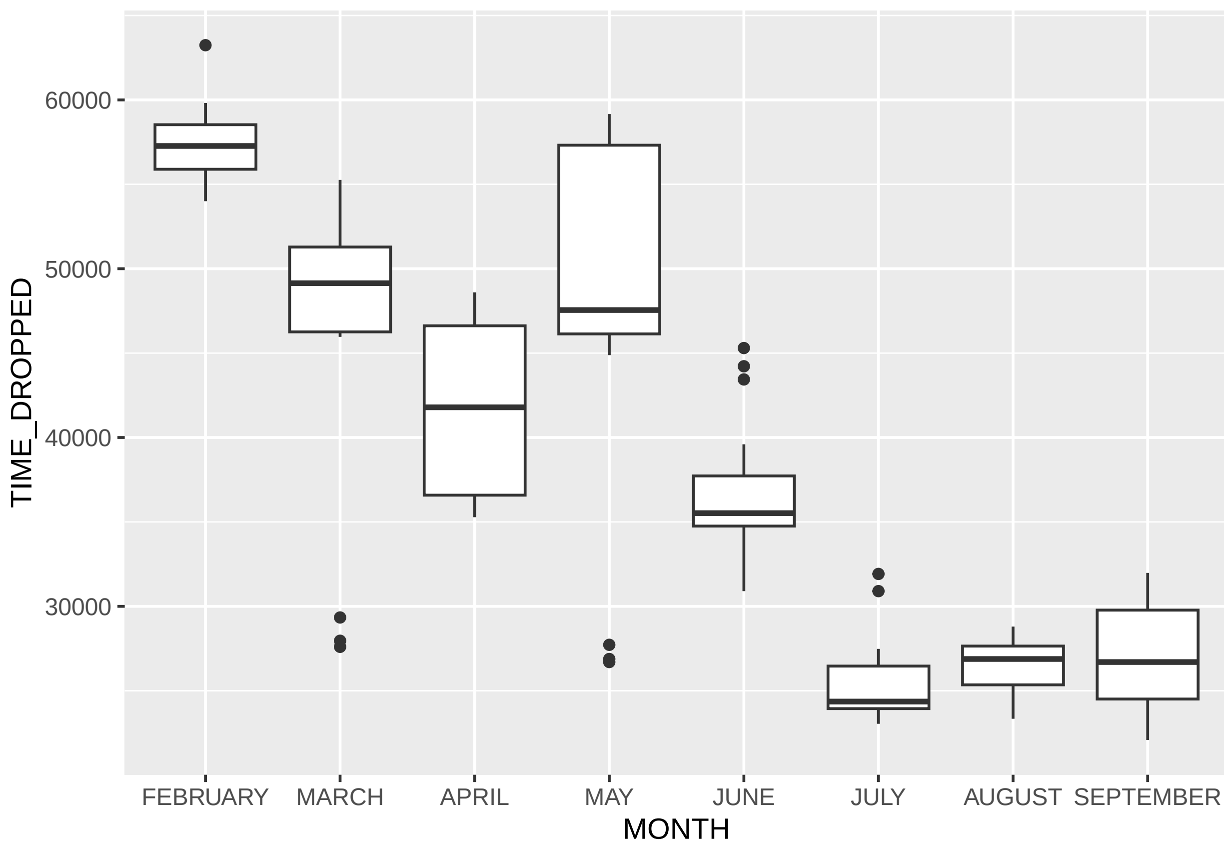


Figure 2: relationship between month and time of day video was taken

4 Modeling Total MaxN

4.1 Steps of my thinking

Ultimately I landed on the model outlined below based on the following decisions:

1. Changed model from poisson to negative binomial because this greatly improved the QQ plot (see figure 3)
2. Increased the k values (from the default of 10) because this increased the k index to be almost 1 for the continuous splines (but none of them are above 1 which apparently is ideal)
3. I included visibility as a spline even though this was considered a corrective measure (higher visibility means you will see more fish). I asked Claude what to do with that and it suggested a shrinkage basis (bs = "ts"). I've also included an offset for total video time as there is a few seconds variation in how long the videos are. Including both of these improved AIC scores compared to not including them
4. Even so, we're not seeing a perfect fit of data. There's an outlier (see figure 4) that is messing with some things. Removing it improves the response vs linear predictor plot but doesn't have an effect on response vs fitted values plot. To be clear, I'm not just going to take outliers out of the data, I just wanted to see what kind of effect that was going to have.
5. Greatly increasing k (>20 for each variable) didn't greatly improve diagnostics. Similar to removing the outlier, it improved the residual vs linear predictor plot but in a similar way to simply removing the outlier
6. I tried a model where there was an interaction between Site Type and Desa, however this had a similar effect on the diagnostics to #3 and #4 above. Result of AIC shows we don't want interaction between desa and site type (2 df higher, 3 pt higher AIC criterion for the interaction model vs non-interaction model)
7. I tested whether or not we should include month or time of day dropped. Including both variables had the lowest DF and AIC score (by a TINY margin), but I still feel like these variables are related so I just included month in this model.

4.2 Where I'm at on this

The model itself seems fine but isn't describing most of the variation (see diagnostics below). Not sure if I'm going to report anything about this because there remains a lot of variation not described by the model.

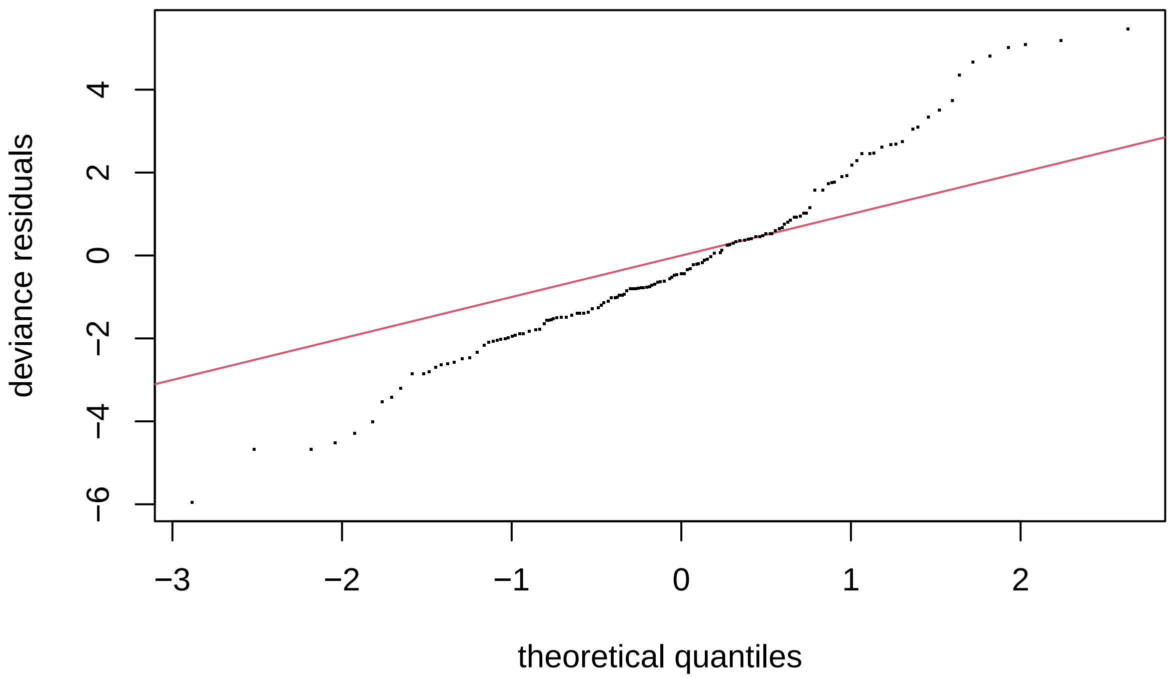


Figure 3: I decided not to use a poisson distribution because the QQ plot looked like this. Chose negative binomial instead which greatly improved the fit (see below for negative binomial QQ plot)

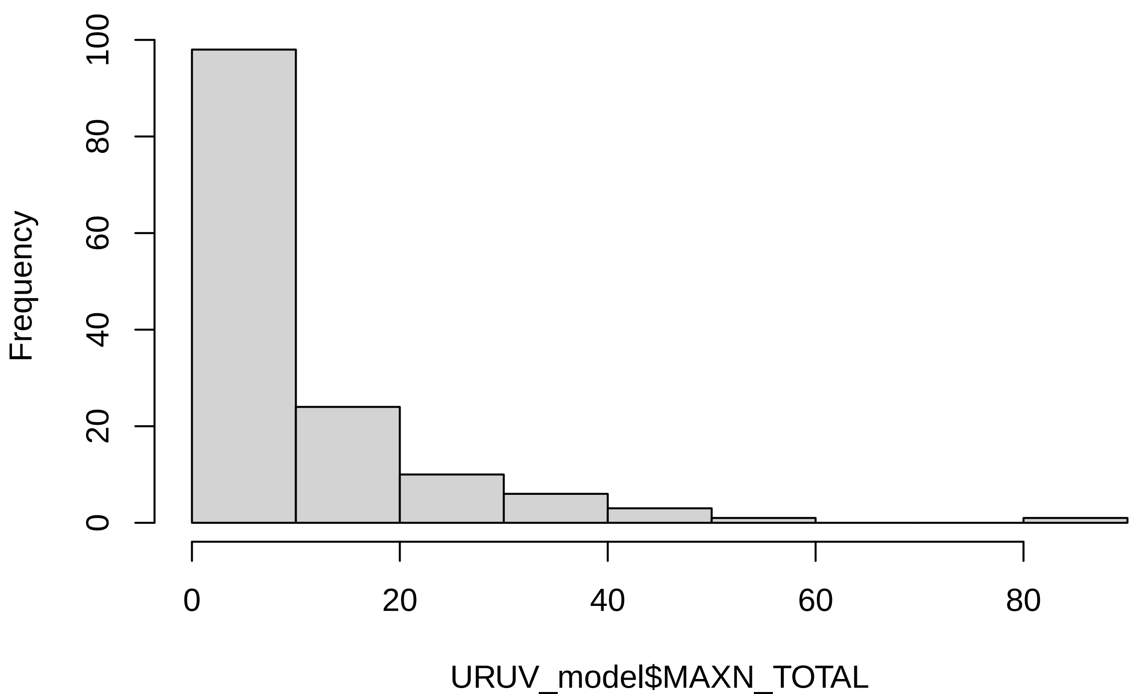
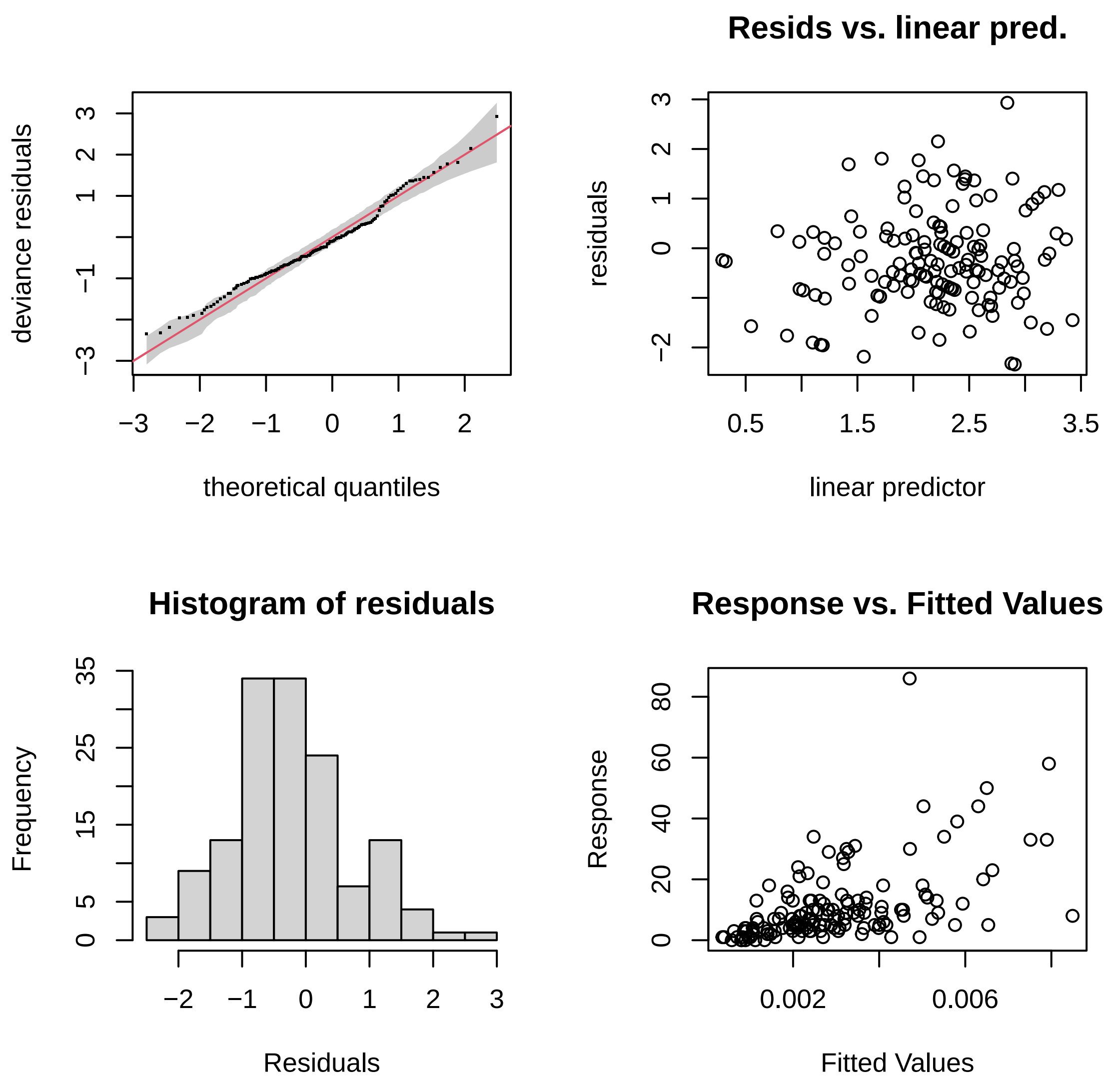


Figure 4: Histogram of Total MaxN results. The outlier is residual vs linear predictor

4.3 Total MaxN model and Diagnostics

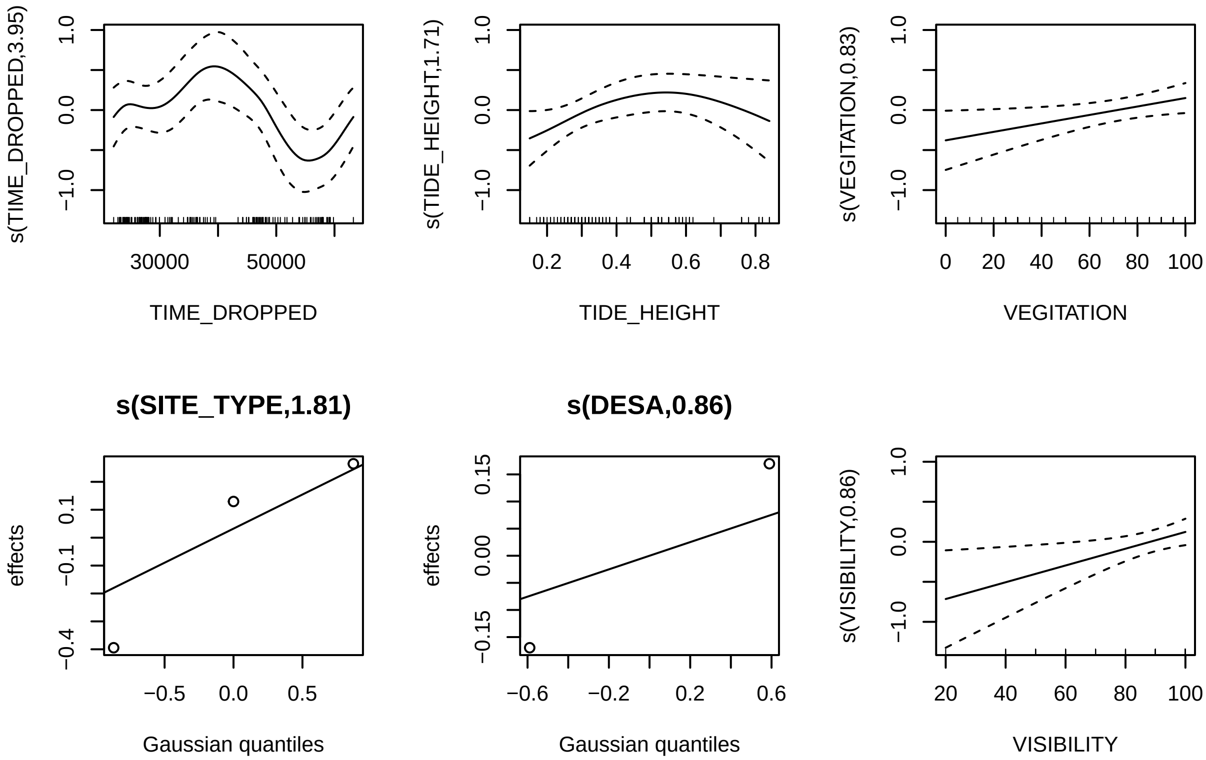
```r
m1.2 <- gam(MAXN_TOTAL ~
              s(TIME_DROPPED, bs = "cc", k = 30) +   # cyclic smooth for time of day
              s(TIDE_HEIGHT, k = 15) +
              s(VEGITATION, k = 15) +
              # Month +
              s(SITE_TYPE, bs = 're') +              # ordered factor
              s(DESA, bs = 're') +  # nominal factor
              s(VISIBILITY, bs = "ts", k = 8)  + # shrinkage smooth - let model determine its importanc
              as.factor(WEATHER_CAT),
            offset = log(TOTALTIME_ANALYSIS), # According to AI: The offset goes inside a log() because
            family = nb(),
            method = "REML",
            data = URUV_model,
            select = TRUE
)

# summary(m1.2)

#gam.check(m1.2, rep = 500) # rep creates confidence intervals on QQ plot

capture.output({
  par(mfrow = c(2, 2)) # Arrange plots in a 2x2 grid
  gam.check(m1.2, rep = 500)
  par(mfrow = c(1, 1)) # Reset graphic parameters
})
```



[1] “ ”
[2] “Method: REML Optimizer: outer newton”
[3] “full convergence after 12 iterations.”
[4] “Gradient range [-0.0002224925,5.7886e-05]”
[5] “(score 470.4263 & scale 1).”
[6] “Hessian positive definite, eigenvalue range [3.385866e-06,45.53882].” [7] “Model rank = 71 / 71”
[8] “ ”
[9] “Basis dimension (k) checking results. Low p-value (k-index<1) may”
[10] “indicate that k is too low, especially if edf is close to k’.”
[11] “ ”
[12] ” k’ edf k-index p-value”
[13] “s(TIME_DROPPED) 28.000 3.955 0.89 0.24”
[14] “s(TIDE_HEIGHT) 14.000 1.713 0.99 0.72”
[15] “s(VEGITATION) 14.000 0.833 0.99 0.71”
[16] “s(SITE_TYPE) 3.000 1.806 NA NA”
[17] “s(DESA) 2.000 0.861 NA NA”
[18] “s(VISIBILITY) 7.000 0.858 0.88 0.20”

```
capture.output({
  par(mfrow = c(2, 3)) # Arrange plots in a 2x2 grid
  plot.gam(m1.2, seWithMean = TRUE)
  par(mfrow = c(1, 1)) # Reset graphic parameters
```

```
})
```



s(TIME_DROPPED,3.95)
TIME_DROPPED
s(TIDE_HEIGHT,1.71)
TIDE_HEIGHT
s(VEGITATION,0.83)
VEGITATION
s(SITE_TYPE,1.81)
effects
Gaussian quantiles
s(DESA,0.86)
effects
Gaussian quantiles
s(VISIBILITY,0.86)
VISIBILITY


```
## character(0)
```

5 Modeling Total Number of species

5.1 Steps of my thinking

I think I'm going through a similar set of steps as total MaxN because both are count data.

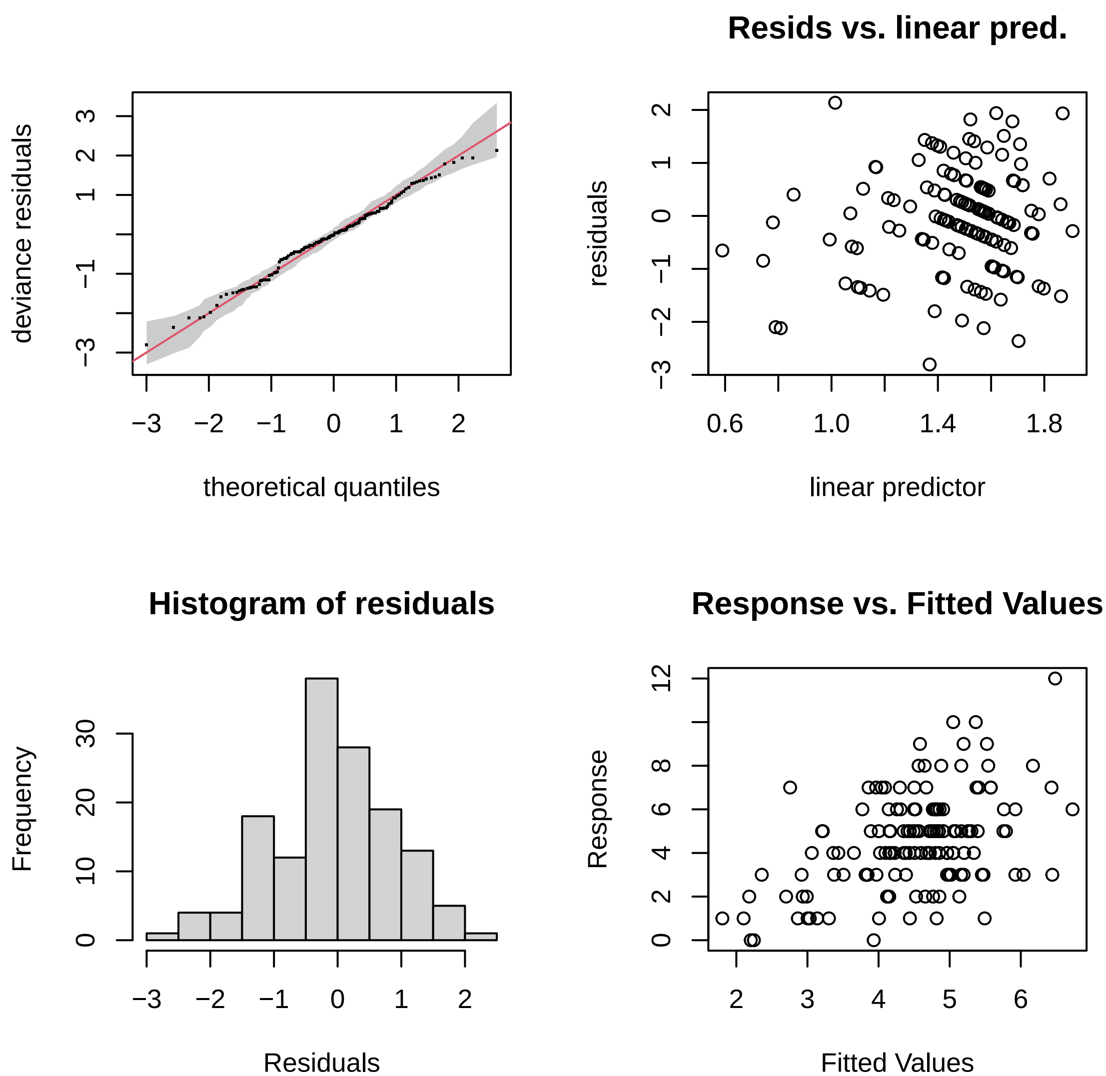
1. Kept poisson dist as QQ plot looks alright and using negative binomial didn't improve anything
2. Ks didn't need much tweakingas most k indices were already above 1
3. Testing using visibility spline and offset of video length. Need to include both of these because doing so results in lowest AIC score but also EDF exceeds k values in gam.check for models that don't have both. However for site type, desa, and visibiltiy, EDF is really low so not sure if there's even an effect there. Even taking out select = TRUE results in really low EDF
4. Including interaction between site type and desa did not improve model
5. I'm using time of day, not month for this model as it improved the AIC score without resulting in almost null EDFs for every variable.

5.2 Where I'm at on this

This feels similar to the first model where the parametrizations themselves seem fine but isn't describing most of the variation (see diagnostics below). Not sure if I'm going to report anything about this because there remains a lot of variation not described by the model.

5.3 Total soecues model and Diagnostics

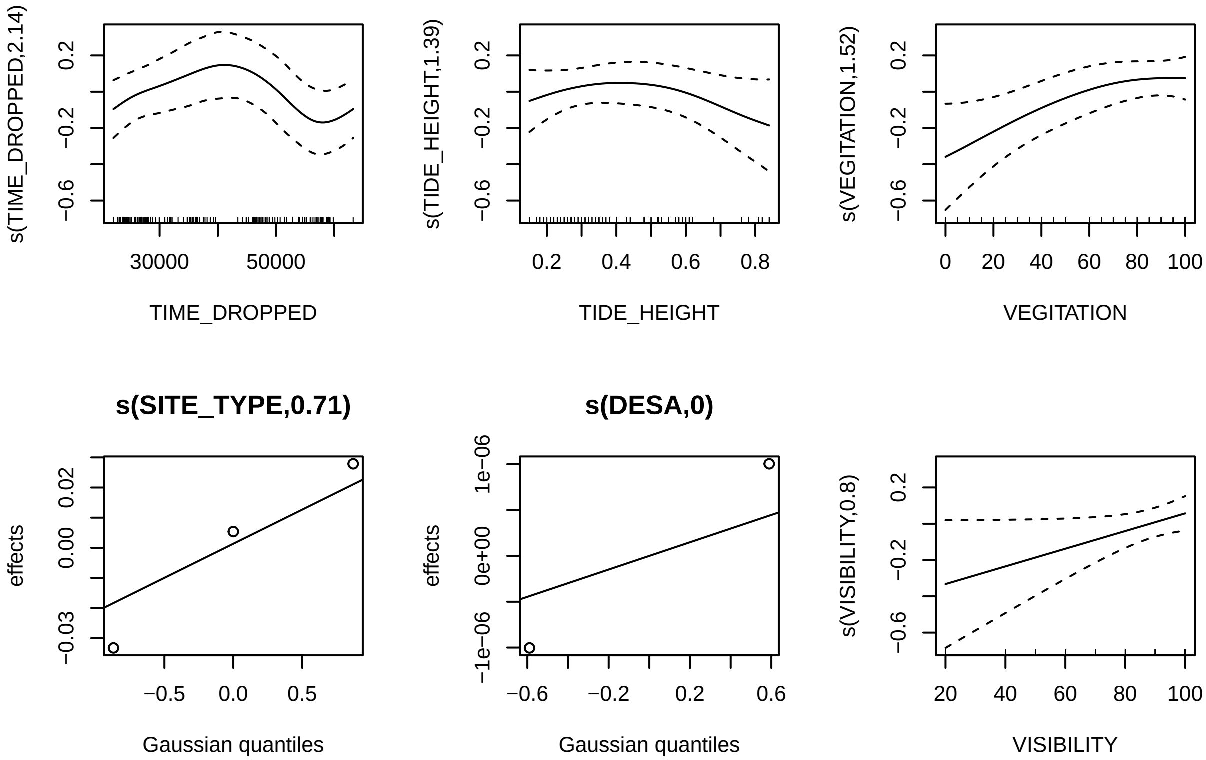
```r
m2.2 <- gam(NO_ALLSPECIES ~
              s(TIME_DROPPED, bs = "cc") +    # cyclic smooth for time of day
              s(TIDE_HEIGHT) +
              s(VEGITATION, k = 15) +
              # MONTH +
              s(SITE_TYPE, bs = 're') +                # ordered factor
              s(DESA, bs = 're') +  # nominal factor
              s(VISIBILITY, bs = "ts", k = 8)  + # shrinkage smooth - let model determine its importanc
              as.factor(WEATHER_CAT),
            offset = log(TOTALTIME_ANALYSIS), # According to AI: The offset goes inside a log() because
            family = poisson(),
            method = "REML",
            data = URUV_model,
            select = TRUE
)

# summary(m1.2)

#gam.check(m1.2, rep = 500) # rep creates confidence intervals on QQ plot

capture.output({
  par(mfrow = c(2, 2)) # Arrange plots in a 2x2 grid
  gam.check(m2.2, rep = 500)
  par(mfrow = c(1, 1)) # Reset graphic parameters
})
```



[1] “ ”
[2] “Method: REML Optimizer: outer newton”
[3] “full convergence after 9 iterations.”
[4] “Gradient range [-6.977396e-05,3.464158e-06]”
[5] “(score 309.1896 & scale 1).”
[6] “Hessian positive definite, eigenvalue range [6.082524e-06,0.3489546].” [7] “Model rank = 46 / 46”
[8] “ ”
[9] “Basis dimension (k) checking results. Low p-value (k-index<1) may”
[10] “indicate that k is too low, especially if edf is close to k’.”
[11] “ ”
[12] ” k’ edf k-index p-value”
[13] “s(TIME_DROPPED) 8.00e+00 2.14e+00 1.10 0.92”
[14] “s(TIDE_HEIGHT) 9.00e+00 1.39e+00 1.04 0.63”
[15] “s(VEGITATION) 1.40e+01 1.52e+00 0.95 0.24”
[16] “s(SITE_TYPE) 3.00e+00 7.06e-01 NA NA”
[17] “s(DESA) 2.00e+00 1.43e-04 NA NA”
[18] “s(VISIBILITY) 7.00e+00 8.01e-01 1.08 0.81”

```
capture.output({
  par(mfrow = c(2, 3)) # Arrange plots in a 2x2 grid
  plot.gam(m2.2, seWithMean = TRUE)
  par(mfrow = c(1, 1)) # Reset graphic parameters
```

```
})
```



s(TIME_DROPPED,2.14)
TIME_DROPPED
s(TIDE_HEIGHT,1.39)
TIDE_HEIGHT
s(VEGITATION,1.52)
VEGITATION
s(SITE_TYPE,0.71)
effects
Gaussian quantiles
s(DESA,0)
effects
Gaussian quantiles
s(VISIBILITY,0.8)
VISIBILITY


```
## character(0)
```

6 Modeling Shannon's Diversity Index

6.1 Steps of my thinking

1. I guess there are two ways to handle using the video time as an offset. I can just divide Shannon by time or I can include as a spline. Comparing those models, it seems like dividing shannon by time is the move. However, every variable has

2. Using month or time of day. To be honest, this seems like the difference is pretty negligible. I'm inclined to use just month (model 4 in 3)

3. Interaction between desa and site type

Table 3: Explanatory variables I'm considering

	df	AIC
m3.1	14.026717	-2105.794
m3.2	9.907819	-2101.253
m3.3	12.844109	-2101.861
m3.4	14.053244	-2105.769